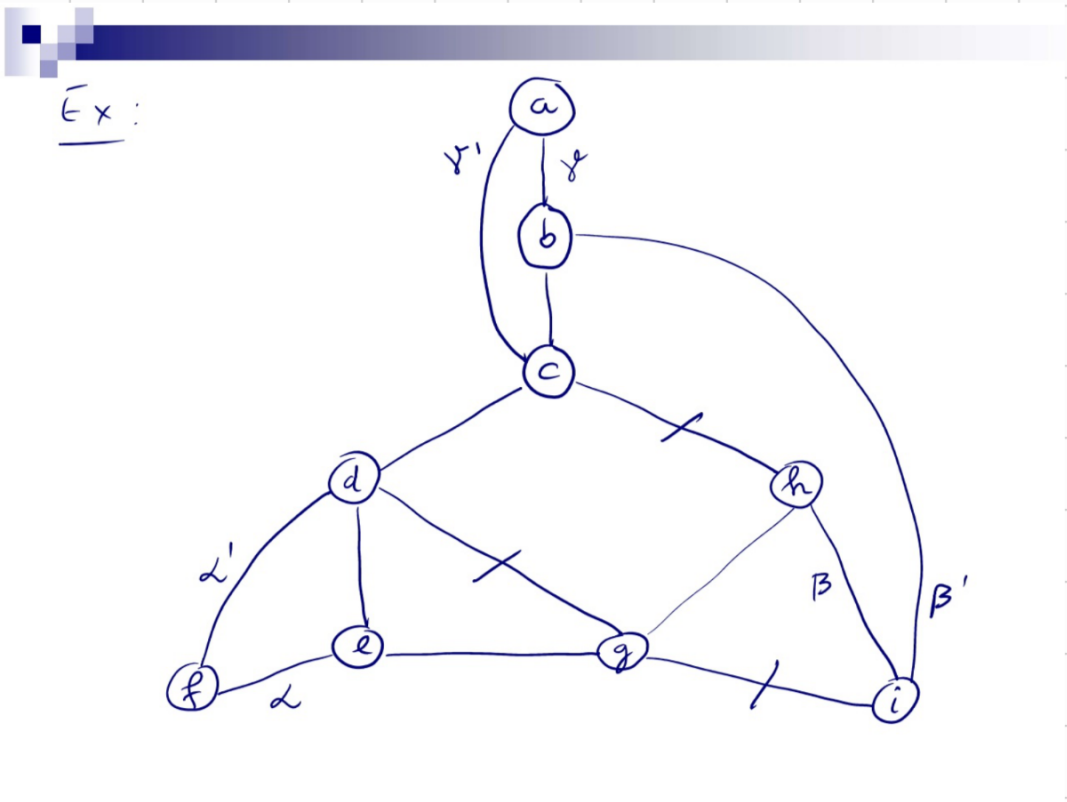
Laborator 10

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2)



$c(a), c(b), \ldots, c(i) = alb$

$\pi(a), \pi(b), \ldots, \pi(i) = null$

$timp = 0$

$E_2 = \emptyset$

explorare(a)

$d(a) = b(a) = b_2(a) = 1$ ($timp = 1$)

$e(a) = (a, a)$

$c(a) =$ cenuşiu

$v \in \{b, c\}$ (succesori a)

$c(b) = alb$ (true)
$\pi(b) = a$
$E_2 = E_2 \cup \{(a,b)\}$

explorare (b)
$d(b) = b(b) = b_2(b) = 2$ (timp = 2)
$e(b) = (b,b)$
$c(b) =$ cenuşiu

$v \in \{a, c, i\} \rightarrow$ succesorii lui b

$c(a) = alb$ (false)
$\pi(b) \neq a$ (false)

$c(c) = alb$ (true)
$\pi(c) = b$
$E_2 = E_2 \cup \{(b,c)\}$

explorare (c)
$d(c) = b_2(c) = b_2(c) = 3$ (timp = 3)
$e(c) = (c,c)$
$c(c) =$ cenuşiu

$v \in \{a, b, d, h\}$

$c(a) = alb$ (false) $\quad //^1 \quad //^3$
$\pi(c) \neq a$ (true), $d(a) < b(c)$ (true)
$b(c) = d(a) = 1$
$e(c) = (c,a)$

c(b) = alb (false)
π(c) ≠ b (false)

c(d) = alb (true)
π(d) = c
$E_2 = E_2 \cup \{(c,d)\}$

explorare (d)
d(d) = b(d) = b_2(d) = 4 (timp = 4)
c(d) = cenușiu
e(d) = (d,d)

$v \in \{c, e, f, g\}$
c(c) = alb (false)
π(d) ≠ c (false)

c(e) = alb (true)
π(e) = d, $E_2 = E_2 \cup \{(d,e)\}$

explorare (e)
d(e) = b(e) = b_2(e) = 5 (timp = 5)
c(e) = cenușiu
e(e) = (e,e)

$v \in \{d, f, g\}$

c(d) = alb (false)
π(e) ≠ d (false)
c(f) = alb (true)

$\pi(f) = e$

$E_2 = E_2 \cup \{(e, f)\}$

explorare (f)

$d(f) = b(f) = b_2(f) = 6$ (timp = 6)

$c(f) =$ cenuşiu

$e(f) = (f, f)$

$v \in \{d, e\}$

$c(d) =$ alb (false)

$\pi(f) \neq d$ (true) $\Rightarrow$ $d(d) < b(f)$ (true)
(4) (6)

$\Rightarrow$ $b(f) = d(d) = 4$

$e(f) = (f, d)$

$c(f) =$ negru

$c(e) =$ alb (false)

$\pi(f) \neq e$ (false)

$b(f) < b(e)$ (true)
(4) (5)

$b(e) = b(f) = 4$

$e(e) = e(f) = (f, d)$

$d(e) < b_2(f)$ (true)
(5) (6)

$E_2 = E_2 \cup \{e(f)\}$ $(f, d) = a'$

Ne oprim.

4) $\delta = \max_{i \in I} \frac{cost(i)}{cost^*(i)}$ (minimizare)

Să se verifice factorul de aproximare

$\delta < 2 - \frac{1}{\lambda}$ pt. arc λ conectat (G)

↓ $\lambda = 2$

$\Rightarrow \delta < 2 - \frac{1}{2} = \frac{3}{2}$

$cost(i) = 11$ (costul soluției găsite – nr. de arce netăiate)

$cost^*(i) = 9$ (costul soluției optime – fiind 9 noduri, acesta este de asemenea și costum optim)

$\Rightarrow \frac{11}{9} < \frac{3}{2}$ (adevărat) $\Rightarrow$ relația este verificată.